

1. I am using 3D slice, I installed needed extensions for plastic surgery. but when I import the MRI file, I can't see 3D model for it.

I'll help you troubleshoot this like a pipeline problem: MRI import → volume visibility → segmentation/model creation → 3D display. The key point is that MRI data usually appears first as **2D slice volume**, not automatically as a 3D surface model.

Thought for 4s

In **3D Slicer**, importing an MRI does **not automatically create a 3D model**. Usually it only loads as a **volume**, shown in the red/yellow/green 2D slice views. To see something in 3D, you need either **Volume Rendering** or **Segmentation** → **3D model**.

Try this first: show MRI as 3D volume

1. Import/load the MRI DICOM correctly:
 - Use **DICOM** module → **Import** → select the folder → **Load**.
 - Make sure you load the correct MRI series, not a scout/localizer image. 3D Slicer's DICOM module is the normal way to import clinical DICOM folders.
2. Go to **Volume Rendering** module.
3. In **Volume**, select your MRI volume.
4. Click the **eye icon** to enable rendering.
5. Try a preset:
 - For MRI, try presets like **MR-Default**, **MR-MIP**, or similar.
 - Adjust **Shift** / opacity if the 3D view looks empty.

Volume rendering displays the image volume directly as a 3D object without segmentation. It does not create a mesh/model yet.

If you need an actual 3D model for plastic surgery

For surgery planning, you probably need a **surface model** of skin, bone, soft tissue, nose, jaw, etc. MRI alone may not automatically give a clean model.

Use:

1. **Segment Editor**
2. Click **Add** to create a segment.
3. Choose your MRI as **Master volume** / source volume.
4. Use tools like:
 - **Threshold**
 - **Paint**
 - **Grow from seeds**
 - **Scissors**

- **Smoothing**
5. Turn on **Show 3D** for the segmentation.

Segment Editor is the module used to define structures of interest in 2D/3D/4D images.

Very important: MRI vs CT

For plastic surgery, especially bone/face/skull planning, **CT is usually much easier than MRI** for generating clean 3D bone models. MRI is better for soft tissue, but it is harder to threshold cleanly because bone is not bright like CT bone.

So if you expect a skull/face bone 3D model and you only imported MRI, that may be the reason you see nothing useful.

Common reasons you see no 3D model

Problem	What to check
You loaded MRI, but no model appears	Normal. MRI loads as volume, not model. Use Volume Rendering or Segment Editor.
3D view is blank	In Volume Rendering, select the correct volume and click the eye icon.
Wrong DICOM series loaded	Load the main axial/sagittal/coronal MRI series, not scout/localizer.
MRI is 2D slices with thick spacing	3D view may look poor/blocky.
You need STL/OBJ model	You must segment first; volume rendering itself is only visualization, not exportable mesh. Slicer forum guidance says volume rendering is display-only; for STL/model export, segment the volume first.

Plastic surgery
extension installed
but no model

Extensions usually help after loading/segmentation; they do not
magically convert every MRI into a clean model.

Best workflow

For quick viewing:

Load MRI → Volume Rendering → eye icon → choose MR preset → adjust opacity

For real surgical 3D model:

Load MRI/CT → Segment Editor → segment anatomy → Show 3D → Export to model/STL

My honest guess: nothing is broken. You imported an MRI volume, but Slicer is waiting for you to either **turn on Volume Rendering** or **create a segmentation** first.

I'll give you a practical "click-by-click" workflow. I'll assume your goal is **bone surface editing for surgical/plastic-surgery planning**, and I'll separate **CT-best workflow** from **MRI fallback**, because that changes the steps a lot.

Thought for 4s

Yes. Here is the detailed workflow. I'll write it for **bone surface editing**, but the same idea works for soft tissue too.

Important: for **bone**, use **CT if you can**. MRI bone segmentation is much harder and may not produce a clean surgical model.

Part 1 — Load the scan correctly

1. Open 3D Slicer

Start 3D Slicer.

2. Import DICOM data

Go to:

DICOM module → Import → choose your DICOM folder → Import

Then select the correct series and click **Load**.

Do not load only a single image if your scan has many slices. You need the full volume series. 3D Slicer's DICOM module is the normal way to import clinical DICOM studies.

3. Check that the volume loaded

You should see the scan in the three 2D slice windows:

View	Meaning
Red	Axial
Yellow	Sagittal
Green	Coronal

Use the mouse wheel or slice slider. If you can scroll through the body/face/skull, your volume loaded correctly.

If you see only one flat image, you may have loaded the wrong series.

Part 2 — Create a bone segmentation

4. Open Segment Editor

Go to:

Segment Editor

Then click:

Add

Name the segment something like:

Bone

Segment Editor is the main Slicer module for defining anatomical structures from 2D/3D medical images.

5. Select the correct source volume

In Segment Editor, check:

Source volume / Master volume

Select your CT or MRI scan.

This is important. If the wrong volume is selected, your painting/thresholding may not work.

Part 3 — Segment bone automatically, if possible

6. Use Threshold

In **Segment Editor**, choose:

Threshold

For **CT bone**, this works well.

Typical CT bone threshold starting range:

Lower: around 150–300 HU

Upper: maximum value

For skull/facial bone, you may need to adjust. Drag the threshold sliders until the bone turns highlighted.

Then click:

Apply

Now Slicer creates a bone mask.

7. Turn on 3D preview

Click:

Show 3D

You should now see a bone-like 3D surface in the 3D view.

At this point, you are seeing a **segmentation surface**, not just volume rendering.

Part 4 — Clean the bone segmentation

The first result is usually messy. You may see floating fragments, scan-table pieces, teeth artifacts, noise, or unwanted anatomy.

8. Remove floating pieces with Islands

In Segment Editor, choose:

Islands

Useful options:

Option	Use
Keep largest island	Keeps the largest connected object
Remove small islands	Removes small fragments
Split islands to segments	Separates disconnected pieces

For skull/facial bone, **Keep largest island** can help, but be careful: teeth, jaw, or thin bone may be disconnected depending on scan quality.

9. Cut unwanted parts with Scissors

Choose:

Scissors

You can use it in:

Mode	Use
2D slice view	Precise slice-by-slice removal
3D view	Fast cutting of large unwanted areas

Set operation to remove/erase, then draw around the part you want to delete.

This is useful for removing:

- neck area
- scan table
- unrelated skull parts
- artifacts
- extra anatomy

10. Manual correction with Paint and Erase

Use:

Paint to add missing bone

Erase to remove wrong areas

For surgery planning, this manual correction step is often unavoidable.

Use these views together:

- axial
- sagittal
- coronal
- 3D view

Do not trust only the 3D view. Always check the 2D slices.

Part 5 — Smooth the bone surface

11. Use Smoothing

In Segment Editor, choose:

Smoothing

Common options:

Smoothing type	Use
Median	Removes small rough noise
Opening	Removes small bumps/fragments

Closing	Fills small holes
Joint smoothing	Smooths boundaries between segments

Start with a small value.

Example:

Kernel size: 1–2 mm

Do not over-smooth if surgical accuracy matters. Too much smoothing can change the real bone shape.

Slicer community guidance commonly recommends smoothing after threshold segmentation to clean rough STL/model output.

Part 6 — Edit the bone surface shape

Now comes the important plastic-surgery part.

What you can edit inside Slicer

In Slicer, you can reasonably do:

- remove bone regions
- trim/cut surfaces
- fill holes
- smooth rough areas
- separate parts
- manually paint/erase anatomy
- prepare STL/OBJ export

What Slicer is not ideal for

Slicer is not the best tool for advanced artistic/engineering mesh editing, such as:

- sculpting a nose bridge
- designing an implant shape
- moving jaw bone segments precisely
- simulating before/after cosmetic results
- making polished surgical implant CAD geometry

For that, use Slicer to create the medical model, then export to another tool like:

- Blender
- Meshmixer
- FreeCAD
- Mimics / 3-matic
- surgical planning software

Slicer is great for **medical segmentation**. Blender/Meshmixer/CAD tools are better for **surface sculpting/design**.

Part 7 — Export the segmentation as STL/OBJ

12. Go to Segmentations module

Go to:

Segmentations

Find your segmentation node.

13. Export to file

Open:

Export to files

Choose:

File format: STL or OBJ

Destination folder: your folder

Then click:

Export

3D Slicer supports exporting segments to STL or OBJ for 3D printing or external visualization/processing.

Use:

Format

Best for

STL	3D printing, surgery model, mesh tools
OBJ	mesh editing, sometimes keeps more surface workflow compatibility

Part 8 — Optional: improve the exported model in Slicer

14. Use Surface Toolbox

If the exported model is rough, open:

Surface Toolbox

This can clean and optimize surface models. It supports operations such as cleaning, smoothing, decimation, and surface processing.

Useful tools:

Tool	Meaning
Clean	Removes unused/degenerate mesh points
Smooth	Makes surface less rough
Decimate	Reduces polygon count
Fill holes	Closes small holes
Normals	Fixes surface direction issues

Be careful with **Decimate**. It makes the model lighter, but too much decimation loses anatomical detail.

Part 9 — If you only have MRI

MRI is harder for bone.

MRI workflow

Use:

Segment Editor → Paint / Draw / Grow from seeds / Threshold if possible

But MRI bone does not usually threshold cleanly like CT.

Try this:

1. Load MRI.
2. Go to **Segment Editor**.
3. Add segment: **Bone candidate**.
4. Try **Threshold**, but expect poor results.
5. Use **Paint** and **Erase** manually.
6. Use **Grow from seeds** if boundaries are visible.
7. Use **Smoothing**.
8. Show 3D.
9. Export STL/OBJ.

For facial bone surgery, I would strongly recommend getting a **CT/CBCT scan** if medically appropriate and available.

Recommended practical workflow for you

For plastic surgery bone surface editing, use this order:

1. Load CT/MRI DICOM
 2. Confirm scan in 2D views
 3. Segment Editor
 4. Add Bone segment
 5. Threshold bone
 6. Apply
 7. Show 3D
 8. Islands: remove noise
 9. Scissors: cut unwanted parts
 10. Paint/Erase: manual correction
 11. Smoothing: clean surface
 12. Segmentations: export STL/OBJ
 13. Surface Toolbox: clean/smooth/decimate if needed
 14. Open STL in Blender/Meshmixer/CAD for advanced surface reshaping
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One safe warning

For real plastic surgery planning, do **not** rely only on your own edited model for medical decisions. Segmentation errors can change anatomy. Use this for learning, visualization, or discussion, but final surgical planning should be reviewed by a qualified clinician or surgical planning specialist.

3D Slicer + Blender is the best low-cost / flexible workflow:

3D Slicer = medical image segmentation

Blender = mesh editing, reshaping, visual mockup

For your goal, use this workflow:

Overall workflow

CT/MRI DICOM

→ 3D Slicer

→ segment bone

→ segment skin/face surface

→ export STL/OBJ

→ Blender

→ reshape bone

→ manually deform/sculpt skin

→ compare before/after

Step 1 — In 3D Slicer: create bone model

Use **CT/CBCT** if possible.

1. Open **3D Slicer**
2. Load DICOM:

DICOM → Import → Load

3. Go to:

Segment Editor

4. Click **Add**
5. Name segment:

Bone

6. Use **Threshold**
 - For CT bone, try lower value around:

150–300 HU
 - Adjust until bone is selected.
7. Click **Apply**
8. Click **Show 3D**
9. Clean it using:
 - **Islands**: remove small floating parts
 - **Scissors**: cut unwanted regions
 - **Paint / Erase**: manual fixes
 - **Smoothing**: smooth rough surface

Now you have a 3D bone segmentation.

Step 2 — In 3D Slicer: create skin/face model

1. In **Segment Editor**, click **Add**
2. Name it:

Skin

3. Use one of these:
 - **Threshold** if the outer surface is clear
 - **Paint / Erase** manually
 - **Grow from Seeds** if available and useful
4. Click **Show 3D**
5. Clean the skin surface with:
 - **Scissors**
 - **Smoothing**
 - **Islands**

Now you should have:

Bone model

Skin/face model

visible together in Slicer.

Step 3 — Export bone and skin

Go to:

Segmentations → Export to files

Export both segments as:

Bone.stl

Skin.stl

or:

Bone.obj

Skin.obj

I recommend **OBJ** if you plan to edit in Blender, and **STL** if you plan to 3D print.

Step 4 — Import into Blender

In Blender:

File → Import → STL/OBJ

Import:

Bone

Skin

Then align/check scale. Usually Slicer exports in **millimeters**, so keep scale consistent.

Step 5 — Reshape bone in Blender

For bone editing:

1. Select the bone mesh
2. Use:
 - **Edit Mode** for vertex/face editing
 - **Sculpt Mode** for smooth reshaping
 - **Grab brush** to move surface
 - **Smooth brush** to smooth
 - **Knife / Bisect** to cut
 - **Boolean modifier** to remove or combine parts

Useful Blender modifiers:

Modifier	Use
Smooth	Make surface smoother
Decimate	Reduce mesh size
Boolean	Cut or combine geometry
Shrinkwrap	Fit one mesh to another
Lattice	Deform larger areas smoothly

Step 6 — Create updated skin/face mockup

This is the tricky part.

Blender will not automatically know how real skin moves after bone surgery. You need to manually deform the skin mesh.

Basic method:

1. Select the **Skin** mesh
2. Duplicate it:

Skin_Before
Skin_After

3. Hide **Skin_Before**
4. Edit **Skin_After**
5. Use **Sculpt Mode**
6. Use:
 - **Grab**
 - **Smooth**
 - **Elastic Deform**
 - **Inflate/Deflate**
7. Move skin surface according to the edited bone shape

So your final scene has:

Original skin
Original bone
Edited bone
Manually edited skin prediction

Best Blender setup

Use separate objects:

Bone_Before
Bone_After
Skin_Before
Skin_After

Then you can compare:

Object	Purpose
Bone_Before	Original anatomy
Bone_After	Your edited bone
Skin_Before	Original face surface
Skin_After	Predicted/manual face result

Important reality check

This workflow is great for:

learning
visualization
rough planning
before/after concept mockup
portfolio/demo images
discussion with doctor/designer

But it is **not a clinically accurate plastic surgery simulator**. The skin result is a manual approximation unless you use advanced soft-tissue simulation.

My recommended next step

Start with this small test:

1. Segment skull/bone in Slicer
2. Export Bone.stl
3. Import Bone.stl into Blender
4. Make one simple reshape

5. Export edited bone
6. Reopen it in Slicer or view it in Blender

After that works, add the skin model.

Do **bone first**, then **skin**. Trying to do both at once will get messy fast.

I don't have the Blender software. how can I download it?
Thought for a couple of seconds

Download Blender from the **official Blender website**:

Official download page:

<https://www.blender.org/download/>

Blender is **free and open source**, and it supports **Windows, macOS, and Linux**.

For your case, choose:

If you use Windows

1. Open the download page.
2. Click **Download Blender**.
3. Choose **Windows Installer**.
4. Run the **.msi** installer.
5. Open Blender after installation.

There is also a **Windows Portable (.zip)** version if you do not want to install it normally.

If you want a stable long-term version

Use the LTS version:

[Blender 4.5 LTS](#)

For your 3D Slicer + Blender workflow, the normal latest version or **4.5 LTS** is fine. I'd choose **Blender 4.5 LTS** for stability.

Once you have **Blender installed**, here's how to see and work with your reshaped 3D model step by step:

1. Open Blender

- Launch Blender.
 - By default, a cube appears in the scene. You can delete it:
 - Select the cube → press **X** → **Delete**.
-

2. Import your model (STL/OBJ from 3D Slicer)

1. Go to the top menu: **File** → **Import** → **STL** (or **OBJ** if you exported as OBJ).
2. Navigate to your exported file(s), e.g. **Bone.stl** or **Skin.stl**.
3. Select it → **Import STL**.

Tip: You can import multiple files at once to have **bone + skin** in the same scene.

3. Adjust the view

- **Middle mouse button (MMB)** → rotate view
- **Shift + MMB** → pan
- **Scroll wheel** → zoom

You should now see your 3D model in the viewport.

4. Check model scale and position

- Select the object: **Right-click** / **Left-click** (depending on your setup)
 - Press **N** → open **Transform panel** → check **Location**, **Rotation**, **Scale**
 - Usually Slicer exports in **millimeters**, so scale should be correct.
-

5. Enable “Solid” or “Material Preview” view

- On top right of 3D viewport, choose viewport shading:
 - **Solid** → default solid color
 - **Material Preview** → see materials and lighting
 - **Rendered** → realistic lighting (optional)
-

6. Display multiple models

- If you imported both **Bone** and **Skin**:
 - Both appear as separate objects.
 - You can **hide/unhide** using the eye icon in **Outliner** panel (top-right).
 - Move or rotate each model:
 - **Select object** → **G** (Grab) → move
 - **R** → rotate
 - **S** → scale
-

7. Select and reshape the model

- **Select the object** you want to edit (e.g., reshaped bone).
 - Switch to **Edit Mode**:
 - Press **Tab** → Edit Mode
 - Here you can manipulate vertices, edges, and faces directly.
 - Or use **Sculpt Mode**:
 - Press **Tab** → **Sculpt Mode**
 - Use brushes like **Grab**, **Smooth**, **Inflate**, **Flatten** to adjust the shape.
 - Adjust brush size with **F** key and strength with **Shift + F**.
-

8. Show before/after comparison

- Duplicate your model:
 - Select → **Shift + D** → duplicate → move aside
 - Keep one original as reference.
 - Reshape the duplicate.
 - You can toggle visibility in the **Outliner** panel.
-

9. Save your Blender project

- **File** → **Save As** → **.blend** file
 - This preserves all your edits for later.
 - Optional: Export the reshaped model for 3D printing or sharing:
 - **File** → **Export** → **STL / OBJ**
-

Quick summary of controls

Action	Shortcut
Rotate view	MMB
Pan view	Shift + MMB
Zoom	Scroll wheel
Select object	Right-click / Left-click
Move object	G
Rotate object	R
Scale object	S
Switch Edit / Object Mode	Tab
Sculpt Mode	Tab → Sculpt Mode

Duplicate

Shift + D

Hide object

Outliner eye icon

If you want, I can make a **step-by-step visual workflow diagram showing 3D Slicer → export → Blender → reshaping → skin overlay**, which makes it easier to follow.

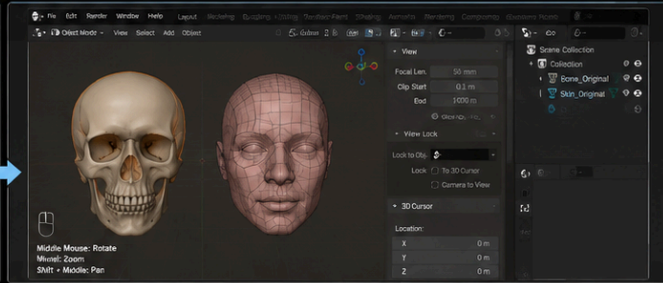
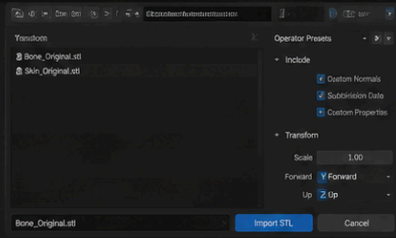
Do you want me to make that?

HOW TO SEE THE RESHAPED MODEL IN BLENDER

Workflow: Import models → Reshape → View the updated result

1 IMPORT YOUR MODELS

1. Open Blender
2. File → Import → STL or OBJ
3. Select your Bone or Skin model
4. Click Import



2 RESHAPE THE MODEL

You can reshape in Edit Mode or Sculpt Mode.

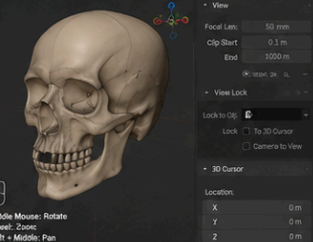
- A. Edit Mode**
- Select the bone
 - Tab → Edit Mode
 - Move vertices (G)
 - Extrude (E)
 - Scale (S), etc.
- B. Sculpt Mode**
- Go to Sculpt Mode
 - Use brushes like Grab, Smooth, Inflate, etc.



Reshape the bone as you need.

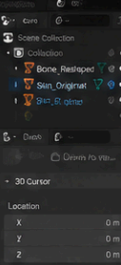
3 CHECK THE RESULT

1. Press Tab → Object Mode
2. You will see the reshaped model
3. Rotate (Middle Mouse), Zoom (Wheel) to inspect it



4 VIEW WITH SKIN / FACE MODEL

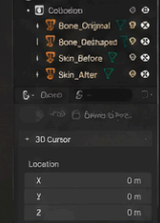
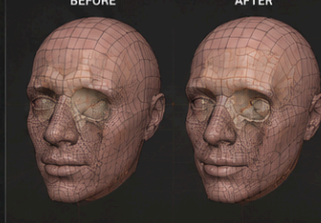
1. Import your skin model (if not imported)
2. Make sure both models are visible in the Outliner
3. You can see how the reshaped bone looks inside the face



5 COMPARE BEFORE / AFTER

Duplicate collections or objects to compare.

- Example:
- Bone_Original
 - Bone_Reshaped
 - Skin_Before
 - Skin_After (edited)
- Use different colors or display modes (Wireframe, Solid, X-Ray) to compare.



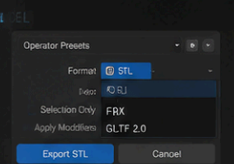
6 RENDER THE RESULT (OPTIONAL)

- Go to Render Properties (camera icon) → Choose Render Engine (Eevee or Cycles) → Adjust lights and materials → Render.



7 EXPORT THE RESHAPED MODEL

1. File → Export → STL / OBJ / FBX to use in other software or 3D printing.



Here's a **step-by-step workflow using InVesalius → Blender** for bone reshaping and visualization of your surgical model:

Step 1 — Segment bone in InVesalius

1. Open **InVesalius**.
2. Load your **CT or MRI DICOM folder**:
 - **File → Open DICOM Folder → Select your scan**
3. Go to **Segmentation** panel.
4. Create a **new segment** for bone:
 - Choose **Threshold** method.
 - Typical CT bone range: ~150–300 HU.
 - Adjust threshold until bone appears highlighted.
5. Optional: Remove small fragments with **Remove Islands**.
6. Optional: Smooth surface with the **Smoothing** function.
7. Verify in 3D preview.

Tip: MRI is harder for bone segmentation; CT is preferred.

Step 2 — Export the segmented bone

1. After segmentation is satisfactory, go to:

File → Export 3D Model

2. Choose **STL** or **OBJ** format.
3. Save to a folder you can access easily.

You now have a 3D bone mesh ready for editing.

Step 3 — Open Blender

1. Install Blender if not already done:
<https://www.blender.org/download/>
2. Open Blender.

3. Delete the default cube: select → **X** → Delete.
-

Step 4 — Import the bone model into Blender

1. **File** → **Import** → **STL** (or OBJ)
 2. Navigate to the exported InVesalius STL file.
 3. Select it → Click **Import STL**.
 4. The bone mesh appears in the 3D viewport.
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Step 5 — Adjust the bone model in Blender

1. Select the bone mesh.
 2. Switch to **Edit Mode** (**Tab**) to move vertices, edges, or faces:
 - **G** → Move
 - **R** → Rotate
 - **S** → Scale
 3. Switch to **Sculpt Mode** for more organic reshaping:
 - Use brushes: **Grab**, **Smooth**, **Inflate/Deflate**
 - Adjust brush size with **F** and strength with **Shift + F**
 4. Duplicate original bone to keep reference: **Shift + D** → move aside
-

Step 6 — Import or create skin/face model (optional)

1. If you want to visualize skin:
 - Segment skin in InVesalius or Slicer.
 - Export as STL/OBJ.
 2. In Blender: **File** → **Import** → **STL/OBJ**
 3. Make both bone and skin visible.
 4. You can **manually deform skin** to match reshaped bone using **Sculpt Mode**.
-

Step 7 — Compare before/after

- Keep copies of original bone/skin as reference.

- Toggle visibility in **Outliner** panel (eye icon).
 - Rotate, zoom, and pan to inspect changes.
-

Step 8 — Export reshaped model

- **File** → **Export** → **STL/OBJ/FBX**
 - Use for:
 - Surgical planning visualization
 - 3D printing
 - Sharing with other software
-

Summary

InVesalius → segment bone → export STL → Blender → reshape bone → optional skin import → manual deformation → export final model

This workflow lets you **edit bone surfaces for plastic surgery planning** using only free tools.

If you want, I can make a **diagram showing InVesalius → Blender workflow** similar to the Slicer+Blender poster you saw before — it makes it very easy to follow visually. Do you want me to make that?

so, what is better in 3D slicer than InVesalius in editing or reshaping bone surface?

Here's a direct comparison between **3D Slicer** and **InVesalius** specifically for **editing/reshaping bone surfaces**:

Feature / Workflow	3D Slicer	InVesalius	Advantage
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Segmentation flexibility	Very flexible. Threshold, manual Paint/Erase, Grow from Seeds, Scissors, Islands, Smoothing, and many extensions.	Limited. Mainly threshold-based segmentation with basic island removal and smoothing.	Slicer: more control for complex bone structures.
3D surface editing	Can view and edit in 3D using Segment Editor → Show 3D , and with extensions like Surface Toolbox or via Blender export for sculpting.	Very limited. You can remove fragments or smooth, but cannot reshape or sculpt bones in detail.	Slicer: supports advanced 3D editing workflow.
Manual corrections	Paint/Erase, Scissors, Islands, smoothing — per slice or in 3D.	Only basic 3D preview and mask corrections.	Slicer: more precise and versatile.
Integration with external editing	Easily exports STL/OBJ to Blender, Meshmixer, or CAD software.	Exports STL/OBJ too, but workflow for detailed reshaping is harder due to fewer segmentation tools.	Slicer: smoother pipeline for reshaping.
Soft tissue / skin overlay	Can segment multiple structures (bone, skin, soft tissue) and show in 3D.	Limited soft tissue segmentation, mostly bone-focused.	Slicer: better for combined visualization.
Plugins / extensions	Many specialized modules (Plastic Surgery, Surface Toolbox, Segment Mesher).	No major extensions for sculpting or advanced editing.	Slicer: more extensible.

Sculpt-like reshaping

Only via **Sculpt Mode in Blender** after export, but easier due to precise segmentation; or using Segment Editor + Surface Toolbox for some direct smoothing/decimation.

Minimal. Must rely on external software after basic export.

Slicer: easier to prepare clean models for reshaping.

Conclusion

- **3D Slicer is better than InVesalius** for bone editing/reshaping because it provides:
 1. Flexible segmentation tools (manual and semi-automatic)
 2. Better 3D visualization and 3D editing preview
 3. More control for cleaning, smoothing, and isolating structures
 4. Easier integration with external sculpting tools like Blender or Meshmixer
 5. Ability to handle multiple anatomical structures simultaneously (bone + skin)
- **InVesalius** is primarily good for **initial bone model extraction** from DICOM data. For detailed reshaping and surgical planning, **3D Slicer** → **Blender** workflow is superior.

If you want, I can make a **side-by-side workflow diagram** showing **Slicer vs InVesalius for bone editing** so you can see exactly why Slicer is preferred for reshaping.

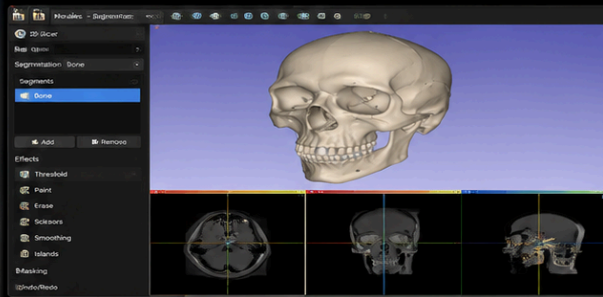
Do you want me to do that?

yes, please

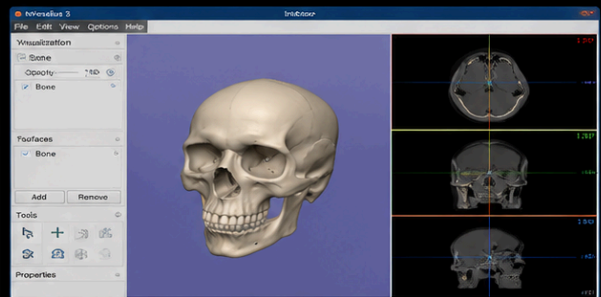
3D Slicer vs InVesalius

Which is better for editing or reshaping bone surface?

3D Slicer (More Powerful for Editing)



InVesalius (More Limited for Editing)



Why 3D Slicer is Better for Editing / Reshaping Bone

	1. Advanced Segment Editor Powerful tools to modify bone precisely. Paint, Erase, Scissors, Grow from Seeds, Islands, Smoothing, Margin, Watershed.
	2. Surface Smoothing & Detail Control Smooth, reduce noise, preserve details, control the surface quality easily.
	3. Local & Global Editing Edit small areas or large regions accurately without affecting other parts.
	4. Smart Tools Islands (remove small parts), Hollow, Shell, Wrap Solid, Margin (thicken or thin surface).
	5. High Quality Surface Generation Produces clean, watertight, high-resolution surface models suitable for further editing in Blender or other software.
	6. Non-Destructive Workflow You can undo/redo, adjust parameters anytime and refine the segmentation iteratively.
	7. Extensible & Customizable Many extensions and modules for advanced workflows.

Best For:

- ✓ Detailed bone segmentation and editing
- ✓ Creating clean models for surgery planning, 3D printing, or simulation
- ✓ Complex anatomical regions (skull, spine, jaw, etc),
- ✓ Users who need precision and flexibility

Limitations of InVesalius for Editing / Reshaping Bone

	1. Basic Editing Tools Limited tools: add, remove, smooth. No advanced options like Scissors, Islands, Grow from Seeds, etc.
	2. Less Precise Control Harder to edit small or complex areas without affecting surrounding regions.
	3. Limited Smoothing Options Smoothing is basic and may cause loss of detail or artifacts.
	4. No Advanced Segmentation Effects Lacks advanced effects like Watershed, Margin, Hollow, Shell.
	5. Surface Quality Output surface may need more cleanup compared to 3D Slicer.
	6. Less Flexible Workflow Fewer options to refine and adjust segmentations iteratively.
	7. Fewer Extensions Limited extensions and community modules compared to 3D Slicer.

Best For:

- ✓ Quick visualization
- ✓ Basic segmentation tasks
- ✓ Users who need simple and fast workflows
- ✓ Not ideal for detailed editing or complex reshaping



Conclusion

If your goal is to edit or reshape bone surface with high precision and control, **3D Slicer is far superior**. InVesalius is suitable for basic tasks but has significant limitations in advanced editing.

Step-by-Step Guide

. Open 3D Slicer

- Launch **3D Slicer** on your computer.
 - Wait until the interface loads.
-

3. Open the DICOM module

- Go to **Modules** → **DICOM** (top-left module menu)
 - Or use the **DICOM icon** on the toolbar.
-

4. Import your DICOM folder

1. Click **Import**.
 2. Browse and select the folder containing your extracted DICOM files.
 3. Click **Import**.
 4. After import finishes, the series will appear in the list of available studies.
-

5. Load the MRI series

- Select the desired series (e.g., “Head MRI T1”)
- Click **Load** → the volume will appear in the red/yellow/green slice views.

Tip: Scroll through slices with the mouse wheel or sliders to verify the volume loaded correctly.

6. Optional — Check 3D

- Go to **Volume Rendering** module
 - Select the loaded volume
 - Enable the **eye icon** to see a 3D preview
 - Adjust preset and opacity to visualize structures
-

7. Segment bone or soft tissue (for jaw/head modeling)

- Go to **Segment Editor** module
 - Click **Add** to create a new segment (e.g., Bone)
 - Select the source volume (your MRI)
 - Use **Threshold**, **Paint/Erase**, or **Grow from Seeds**
 - Click **Show 3D** to see the segmentation surface
 - Refine with **Smoothing**, **Islands**, and **Scissors** if needed
-

8. Export 3D model for further editing

- Go to **Segmentations** → **Export to files**
 - Choose **STL** or **OBJ**
 - Save for import into **Blender** or other 3D software
-

 **Tip:** If you want to practice immediately without downloading large datasets, Slicer has built-in **Sample Data** → **MRHead**, which is a ready-to-use head MRI volume.

I can also make a **visual diagram showing the import → segmentation → export workflow** in 3D Slicer for MRI head/jaw, similar to the previous posters we did.

Do you want me to make that?

To cut part of the **jaw bone** in 3D Slicer, you can use the **Segment Editor** → **Scissors** tool. Here's a step-by-step guide:

1. Load your MRI/CT and segment the jaw

1. Open your volume in 3D Slicer (you've already imported your DICOM).
 2. Go to **Segment Editor**.
 3. Click **Add** to create a segment (name it **Jaw**).
 4. Use **Threshold** to isolate bone (CT preferred; MRI possible but harder).
 5. Click **Show 3D** to visualize the jaw bone.
-

2. Use the Scissors tool to cut

1. In **Segment Editor**, select the segment (**Jaw**).
 2. Choose the **Scissors** effect.
 3. Options:
 - **Shape:**
 - **Brush:** freehand cutting in 2D slices
 - **Circle / Polygon:** draw precise cutting areas
 - **Plane / Box:** cut along a plane in 3D
 - **Operation:**
 - **Erase inside:** removes material inside the drawn shape
 - **Erase outside:** removes material outside the shape
 4. Use **Slice views** (Red, Yellow, Green) to define where you want to cut.
 5. Adjust the **Size / Depth** if using brush or circle.
-

3. Check 3D preview

- Click **Show 3D** to see the jaw after cutting.
 - Rotate/zoom to inspect your cut.
 - If needed, repeat in other slices or planes.
-

4. Clean up

- Use **Smoothing** to smooth cut edges.
 - Use **Islands** to remove small fragments that got detached.
-

5. Export the cut jaw

1. Go to **Segmentations** → **Export to files**.
 2. Choose **STL/OBJ** for 3D editing or 3D printing.
-

 **Tip:** For more precise surgical planning:

- Use **Plane mode** in Scissors for straight cuts.
- You can cut in multiple planes to isolate the jaw section you need.

- After cutting, you can export and further reshape in **Blender** for more complex edits or simulations.
-

If you want, I can make a **visual step-by-step diagram showing how to cut the jaw in 3D Slicer**. This is usually easier to follow than text instructions. Do you want me to make that?

How can I find the buttons or options for the following 1. In Segment Editor, select the segment (Jaw). 2. Choose the Scissors effect.

Thought for a few seconds

Here's exactly where to find them in **3D Slicer**.

1. Open Segment Editor

In the top module dropdown, choose:

Modules dropdown → **Segmentation** → **Segment Editor**

Or search for it:

Module search box → type **"Segment Editor"**

You should now see the **Segment Editor panel** on the left side.

2. Select the segment named "Jaw"

In the left panel, look for the section called **Segmentation**.

You should see a list/table of segments, for example:

Bone

Jaw

Skin

Click the row named **Jaw**.

That row becomes highlighted. That means your **Jaw segment** is selected.

If you do not see "Jaw," then you need to create it first:

1. Click **Add**

2. A new segment appears
3. Rename it to **Jaw**
4. Use **Threshold / Paint / Scissors / other tools** on that selected segment

3. Find the Scissors effect

Still in **Segment Editor**, below the segment list, you should see many effect buttons/icons, such as:

Paint

Erase

Draw

Threshold

Scissors

Smoothing

Islands

Click **Scissors**.

The official Slicer docs say the **Scissors effect works in both slice views and 3D views**, and can erase or draw depending on the operation setting.

4. If you cannot see Scissors

Try these:

A. Make the Segment Editor panel wider

Sometimes the effect buttons are hidden if the left panel is too narrow. Drag the vertical separator to make the left panel wider.

B. Scroll down inside Segment Editor

The effects list may be lower in the panel.

C. Make sure you are really in Segment Editor

Not **Segmentations**, not **Volume Rendering**. The module name at the top should say:

Segment Editor

D. Make sure a segmentation exists

Scissors works on a **segmentation**, not directly on volume rendering. If you only see a 3D volume rendering, Scissors will not cut it. You need to first create a segment from the volume data, then cut that segment.

5. After clicking Scissors

You will see Scissors options like:

Operation

Shape

Slice cut

For cutting jaw bone, start with:

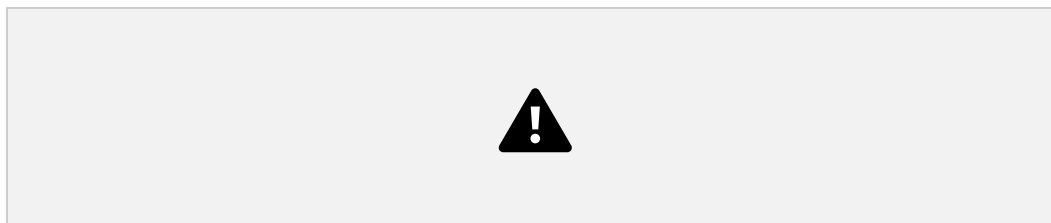
Operation: Erase inside

Shape: Free-form / Rectangle / Circle

Slice cut: Unlimited or Symmetric

Then draw on the jaw area in a **2D slice view** or directly in the **3D view**.

My recommendation: start cutting in the **2D slice views** first. It is much easier to control than cutting directly in 3D.



I can't see the Jaw

From the screenshot you uploaded, it looks like you don't have a segment named "Jaw" yet. In **Segment Editor**, the drop-down shows options like:

Segmentation

Rename current Segmentation

Create new Segmentation

Create new Segmentation as...

Delete current Segmentation

This means you have **no existing segment called “Jaw”**, which is why you can’t select it for Scissors.

How to create the “Jaw” segment

1. In **Segment Editor**, click **Add** (or “Create new Segmentation” if nothing exists).
 2. A new segment appears in the list.
 3. Rename it to **Jaw**:
 - Click the segment name → type “Jaw” → press Enter.
 4. Select the **Jaw** segment. Now the segment is active.
-

Next steps

- After selecting the **Jaw** segment, the **Scissors effect** button should now work.
 - You can then choose **Erase inside** or **Erase outside** to cut the jaw in 2D or 3D views.
-

✓ Summary

- You cannot cut a segment that doesn’t exist.
- First: **Create a segment** → **Rename it to Jaw** → **Select it** → **Use Scissors**.

Do you want me to give a **step-by-step visual guide showing how to create the Jaw segment and cut it** in 3D Slicer?

How to find “Jaw” and the Scissors effect in 3D Slicer

A. Find and select the “Jaw” segment

1 Open Segment Editor

Modules: **Segment Editor**

Segmentation: Segmentation

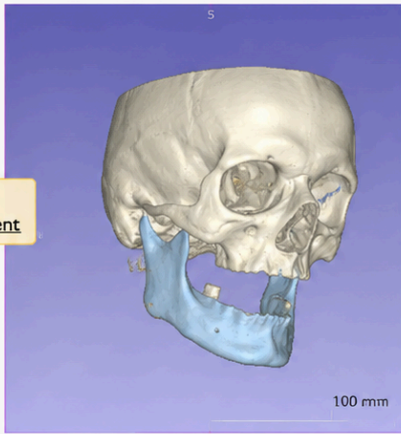
▼ Segmentation

master volume: CTHead

Visible	Color	Name
☐	Green	Bone
☒	Blue	Jaw
☐	Red	Teeth
☐	Yellow	Airways
☐	Blue	Other

Effects

Paint Erase Draw Fill Threshold Scissors



100 mm

1. If you don't see “Jaw”

You need to create it first.

1. Click **Add** +
2. A new segment appears
3. Rename it to **“Jaw”**
4. Then you can edit it

Why you don't see “Jaw”?

- No segmentation exists yet, or
- The segmentation you loaded doesn't contain a segment named “Jaw”.
- You may need to create it.

Make sure you are in “Segment Editor” module (not “Segmentations”). Scissors works on segments, not on the volume directly.

B. Choose the Scissors effect

Effective set: All effects

Effects

Paint Erase Draw Fill Threshold Scissors

Margin Hollow Smoothing Islands

Scissors

Operation: **Erase inside**

Shape: ☒ Free-form ☐ Rectangle ☐ Circle

Slice cut: ☒ Unlimited ☐ Symmetric

☐ Snap to background

1 Click “Scissors” effect

The Scissors options will appear.

2 Set the options (recommended for cutting jaw)

- **Operation:** Erase inside
- **Shape:** Free-form (draw with mouse) or Rectangle / Circle
- **Slice cut:** Unlimited or Symmetric

3 Draw on the jaw in 2D slice views or in 3D view to cut

Start in 2D slice views for more precision.

★ Tip: If you cannot see the effects, make the left panel wider or scroll down inside the Segment Editor panel.